

Week 1

Lesson 1

Introduction to Big Data

Topic

1

What is Big Data?

Topic 1

1-1. Big Data is Everywhere!

- Big data in the movies



Intro to Big Data

Lesson 01

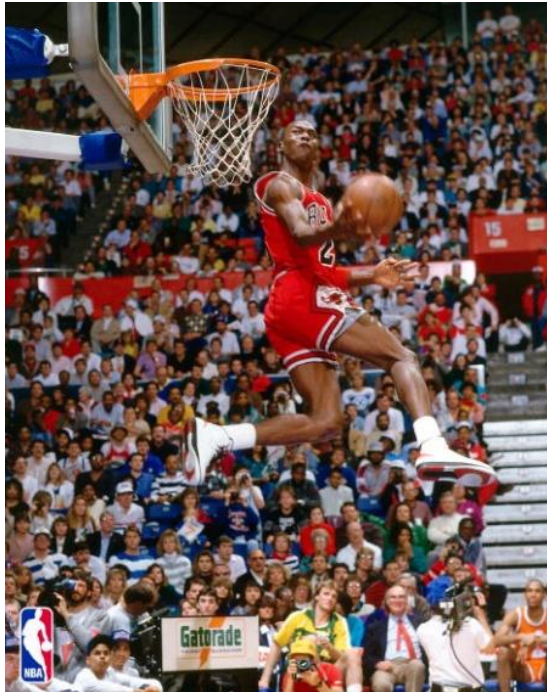


Introduction to Big Data

Topic 1

1-1. Big Data is Everywhere!

- Big data in the sports



Intro to Big Data

Lesson 01



Introduction to Big Data



- **Big data in hospitals**

- ✓ What would be volume and financial impact be if we were to hire another cardiovascular surgeon?
- ✓ What are re-admission patterns for heart failure patients?
- ✓ For a specific diagnosis, what are core interventions that improves the outcomes?



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- **Big data in industries**

Google	predict geographical epidemic flu (trajectory) via search engine log analysis
Google	real time road traffic service
POSCO	determine the purchase time and price of raw materials
Watcha	movie recommendation via taste analysis (no 1., larger than Naver movies)

- **It 1)** prompts response to commercial condition changes,
 - 2) improves credibility and image,
 - 3) reduces expenses,
 - 4) improves productivity,
 - 5) and facilitates administration



Extremely large data
(Wikipedia, Mckinsey)

- Too large to store, manage, and analyze in existing ways using existing storage and existing DBMS SWs

Government and
Industry

- Information technology to predict trends and respond proactively
- Technology to collect, store, manage, search, and analyze large scale data

Before 1600

empirical science

1600–1950s

theoretical science

1950s–1990s

computational science

1990–now

data science

- Each discipline has grown a *theoretical* component.

- Over the last 50 years, most disciplines have grown a third, *computational* branch(e.g. empirical, theoretical, and computational ecology, or physics, or linguistics.)

- The flood of data from new scientific instruments and simulations
- The ability to economically store and manage petabytes of data online

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Topic 1

1-4. Big Data History

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Introduction to Big Data



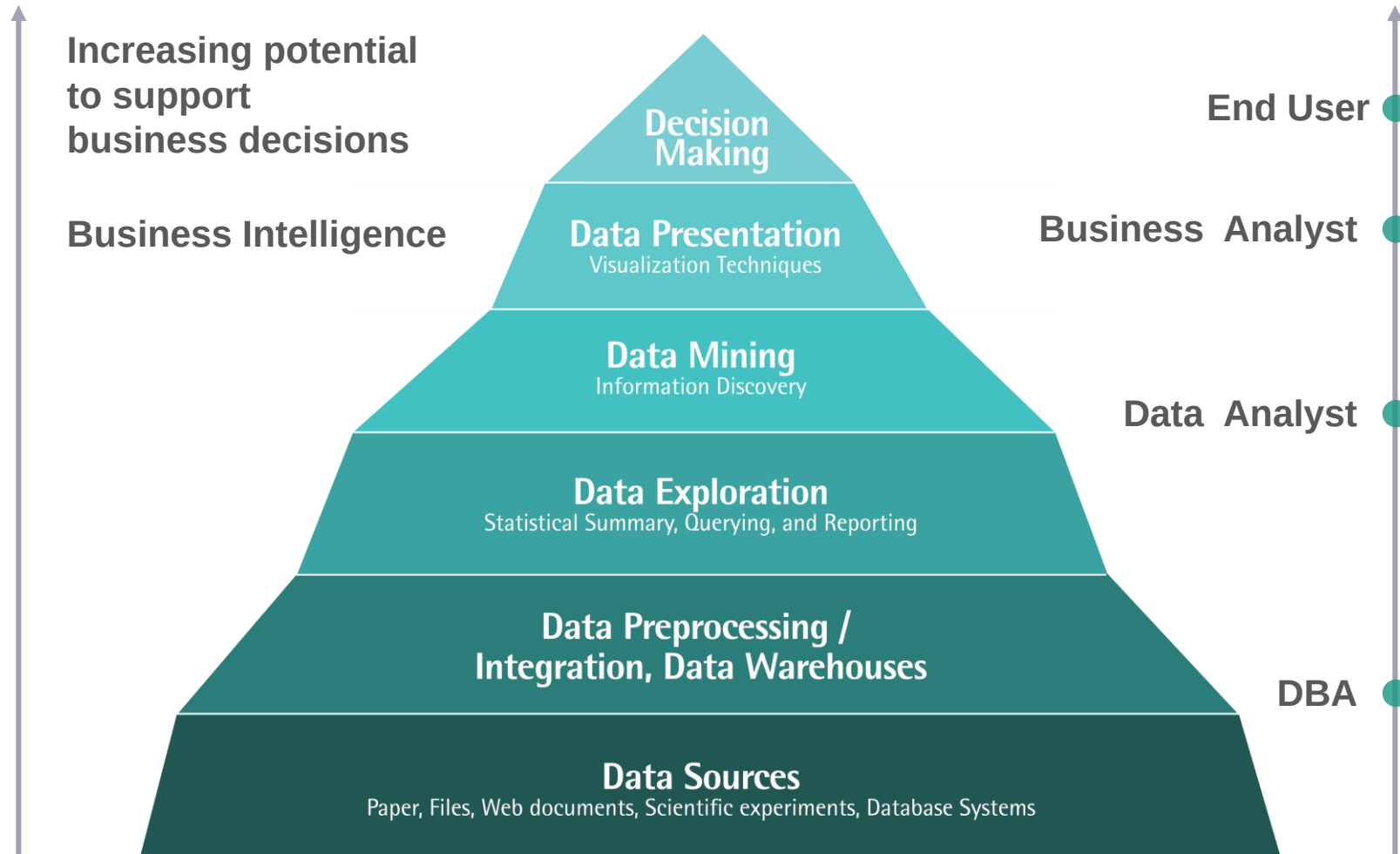
Topic

2

Prospects of Big Data

Topic 2

2-1. Business Intelligence



Intro to Big Data

Lesson 01



Introduction to Big Data



Data Analysts and Data Base
Administrators are promising job!

Every Business Needs DA or DBA



2. Michel J-P, Shen YK, Aiden AP, Veres A, Gray MK, et al. (2011) ***Quantitative analysis of culture using millions of digitized books.*** Science 331: 176–182. doi: 10.1126/science.1199644. Find this article online

3. Lieberman E, Michel J-P, Jackson J, Tang T, Nowak MA (2007) ***Quantifying the evolutionary dynamics of language.*** Nature 449: 713– 716. doi: 10.1038/nature06137. Find this article online

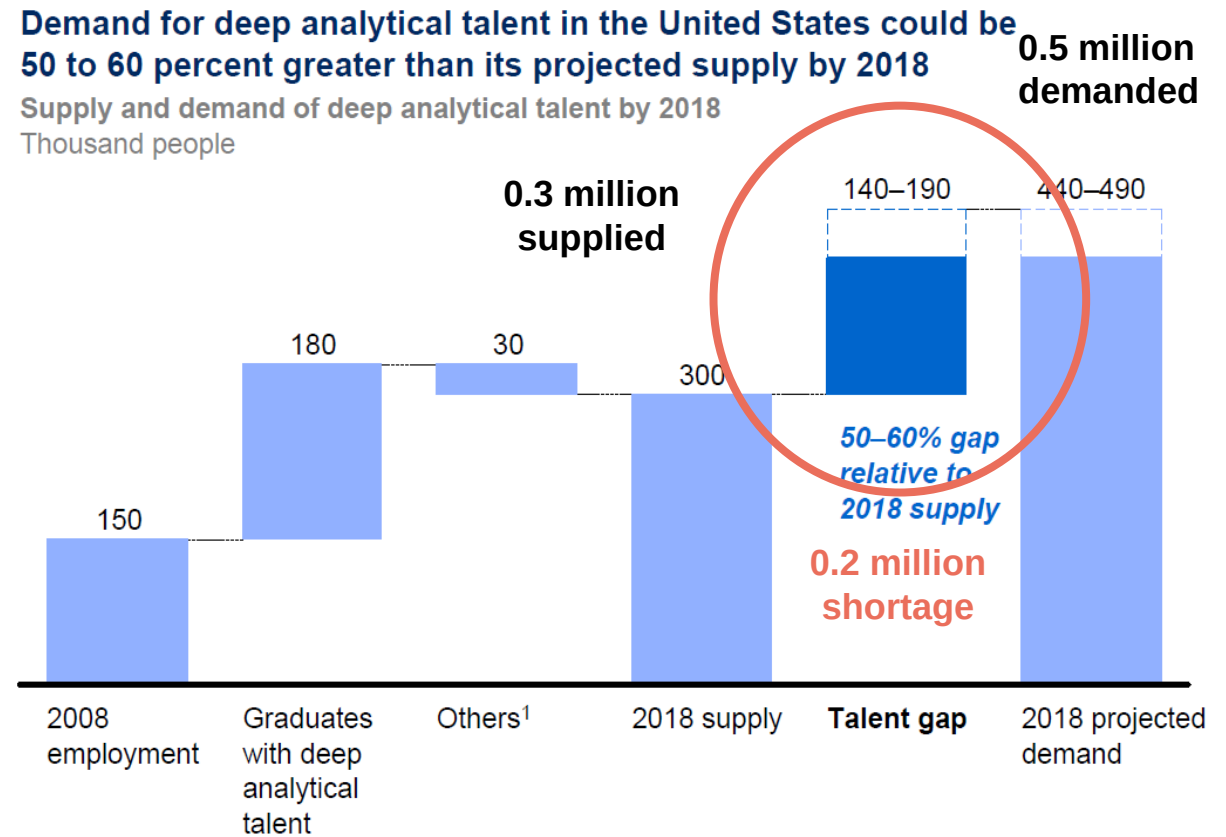
4. Pagel M, Atkinson QD, Meade A (2007) ***Frequency of word-use predicts rates of lexical evolution throughout Indo-European history.*** Nature 449: 717–720. doi: 10.1038/nature06176. Find this article online

6. DeWall CN, Pond RS Jr, Campbell WK, Twenge JM (2011) ***Tuning in to Psychological Change: Linguistic Markers of Psychological Traits and Emotions Over Time in Popular U.S. Song Lyrics.*** Psychology of Aesthetics, Creativity and the Arts 5: 200–207. doi: 10.1037/a0023195. Find this article online

All sciences are data sciences!

- McKinsey Report, 2011

- ✓ predicts rapid growth of Big data industry
- ✓ In USA at 2018, 0.5 million big data experts demanded but only 0.3 million supplied

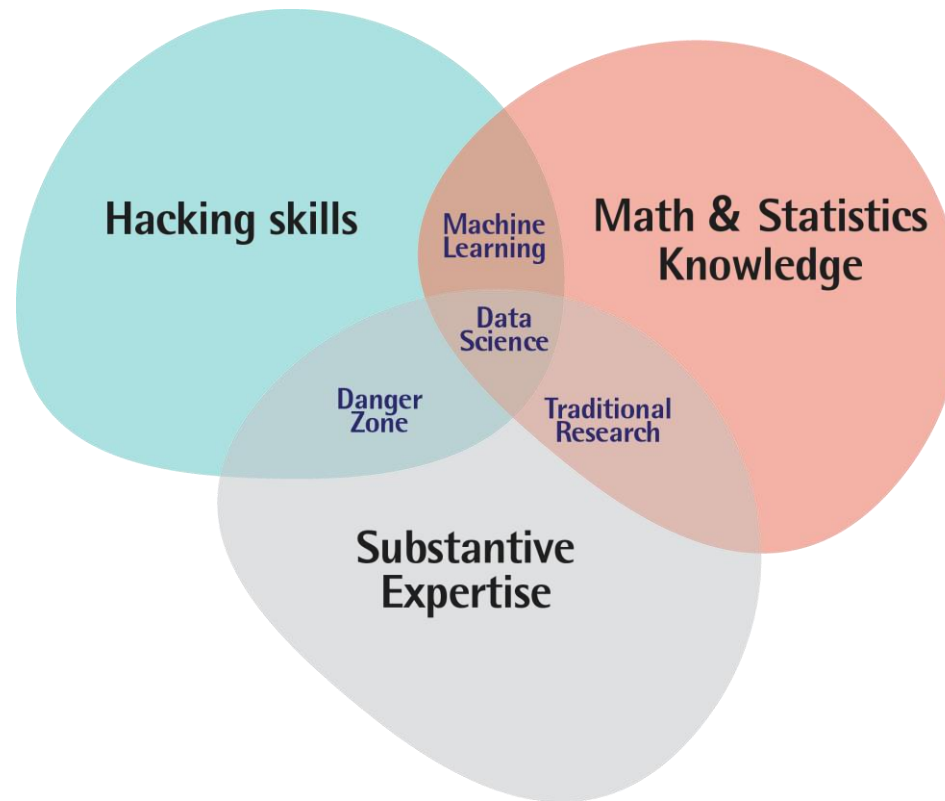


¹ Other supply drivers include attrition (-), immigration (+), and reemploying previously unemployed deep analytical talent (+).
SOURCE: US Bureau of Labor Statistics; US Census; Dun & Bradstreet; company interviews; McKinsey Global Institute analysis

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- Drew Conway's data science venn diagram





- ✓ If you're a **DBA**, you need to learn to deal with unstructured data
- ✓ If you're a **statistician**, you need to learn to deal with data that does not fit in memory
- ✓ If you're a **software engineer**, you need to learn statistical modeling and how to communicate results
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